



An Interactive Guided Meditation Web Experience

Team Members: Aldo Pinelli, Hamin Jang, Stefano Morano, Alessandro Cupini

Abstract

IN-Motion is an interactive guided meditation experience hosted on the web. It combines voice, music, and a dynamic full-screen particle scene, personalized in real time using the user's personal story and real-time facial expression reading. Unlike traditional apps where users operate digital controls, IN-Motion offers a natural interface: users close or open their eyes, speak, and look at the screen. In response, the system generates guidance, music, and adaptive visuals derived from what was spoken and how their facial expressions reacted. The system helps users improve their personal meditation practices through artificial intelligence and generative music models. Each user maintains a dedicated profile, accessible across devices, to record meditation sessions and track emotional progress over time.

1 Overview & System Architecture

The goal of **IN-Motion** is to transform passive meditation into an active, responsive journey. By capturing both implicit biological signals (facial expressions) and explicit input (spoken narratives), the system provides continuous multi-sensory feedback through:

- **Generative Audio & Voice:** Dynamic background music and context-aware spoken guidance.
- **Adaptive Visuals:** Real-time particle visual rendering that morphs based on emotional states.
- **User Progression Tracking:** Cross-device user profiles that log emotion analytics across sessions.

2 Challenges & Technical Solutions

During the development of this web application, a key challenge was determining how to reliably classify human emotions:

- **Sensor Integration Trade-offs:** Initial attempts focused on integrating external hardware sensors (e.g., biometric wearables). However, these required complex hardware setups without adding proportionate value to the computational goals.
- **Facial Emotion Recognition:** To eliminate extra hardware burdens, the system adopts computer-vision-based facial emotion recognition leveraging *Russell's Circumplex Model of Affect*.
- **AI Synthesis with LLMs:** Facial affect data is combined with user storytelling. These inputs are aggregated and processed by **Claude** (Anthropic Large Language Model) to construct real-time adaptive narrative responses.

- **Applying the iso-principle around MuseMorphose::** valence is handled by choosing a reference track which matches the initial emotional state, arousal is dynamically mapped, bar by bar, to rhythmic intensity and polyphony as the piece descends toward Q4. MuseMorphose outputs a single-track piano line, so voices had to be split and orchestrated to match a meditative atmosphere.
- **Tracking eye closure:** we implemented the "Eye Opening Ratio" calculation (the ratio between eye height and width based on 6 key points) and established an empirical blink threshold calibrated to our real lighting conditions.

3 Technologies Used

Component	Technology / Tool
Backend, Music & Particles	Python (Core Logic, Music Generation, Particle Computations)
Frontend Web Application	JavaScript (Client-side interactive interface)
Facial Recognition	MediaPipe (Real-time face tracking and emotion metrics)
Text-to-Speech (TTS)	ElevenLabs (Natural voice output generation)
Speech-to-Text (STT)	OpenAI Whisper (Voice input transcription)
Server Infrastructure	UVicorn (ASGI web server supporting Python API calls & models)

4 Team Members & Roles

- **Aldo Pinelli:** Particle rendering engine, calculation of particle dispositions, finite-state machine (automa) for experience staging.
- **Hamin Jang:** Generative music engine integration and speech-to-text pipeline.
- **Stefano Morano:** Web application porting, server implementation, user authentication, and database architecture.
- **Alessandro Cupini:** Visual arts design and text-to-speech engine integration.

5 Project Resources

- **GitHub Repository:** <https://github.com/stefano-morano/IN-Motion>
- **Demo Video:** <https://youtu.be/bLn7dLulFYo>